

They Lurk Around the Corner: Emerging Infections

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More national and global resources are being mobilized to identify, surveil, and combat the new, continually emerging infectious threats. The foundation for such complex response lies in research aimed at better understanding the agents that cause infectious diseases. The University of Iowa's Center for Emerging Infectious Diseases, based in the College of Public Health, is collaborating with national and international experts in adapting and developing laboratory capabilities to detect and study emerging pathogens.

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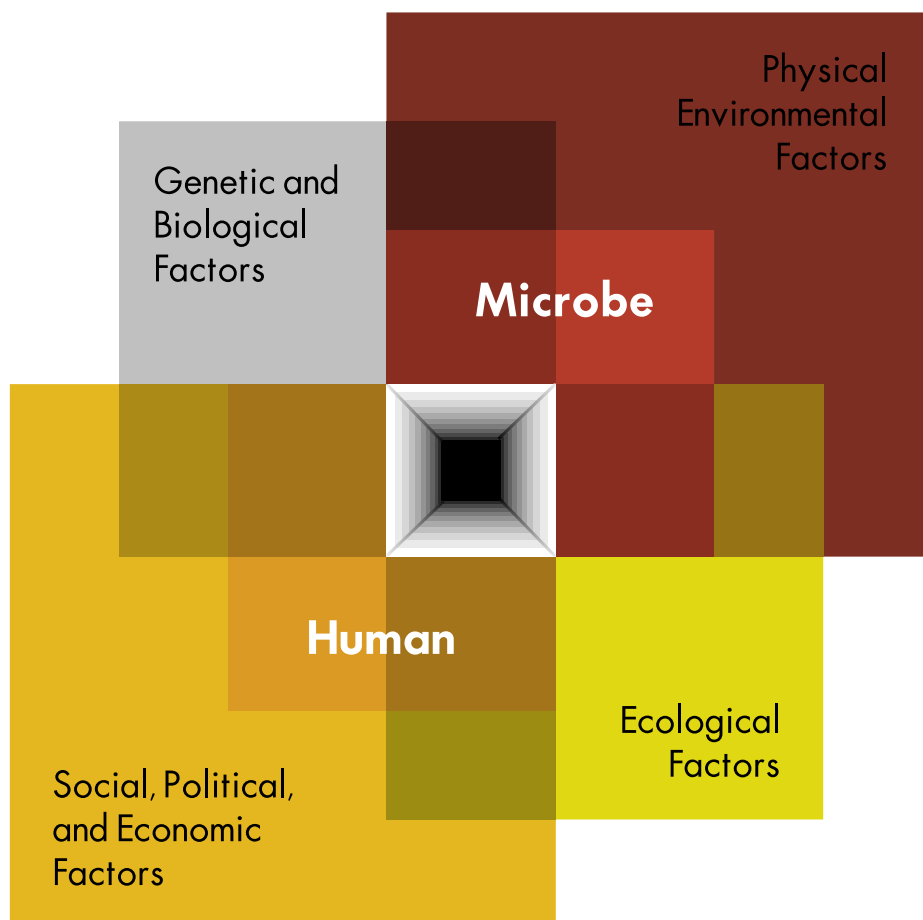
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COVER FIGURE. The Convergence Model. At the center of the model is a box representing the convergence of factors leading to the emergence of an infectious disease. The interior of the box is a gradient flowing from white to black; the white outer edges represent what is known about the factors in emergence, and the black center represents the unknown (similar to the theoretical construct of the “black box” with its unknown constituents and means of operation). Interlocking with the center box are the two focal players in a microbial threat to health—the human and the microbe. The microbe-host interaction is influenced by the interlocking domains of the determinants of the emergence of infection: genetic and biological factors; physical environmental factors; ecological factors; and social, political, and economic factors.

HISTORY

During the 1970s, while many experts believed that victory over infectious diseases was fully at hand, HIV and a number of other new infectious diseases were clandestinely invading human populations. Unaware of this phenomenon, during the same decade, the U.S. Surgeon General and others influenced the shift in public health away from infectious diseases to chronic illnesses. Subsequently, much of our national resources to prevent and control infectious agents quickly eroded. It wasn't until the early 1990s that U.S. public health officials, now aware of the magnitude of the problem, realized their mistake and began calling for a return of public health's focus upon infectious disease prevention. Recent events, such as acts of bioterrorism and the SARS epidemic, have amplified this call. Many policy makers believe that even more resources—national as well as global—should be mobilized to identify, surveil, and combat the new, continually emerging infectious threats.

The lesson brought home by HIV and the subsequent smaller but no less deadly epidemics caused by previously unknown microbes is a lesson of complacency. Humankind remains ignorant of the full range of diseases caused by microbes, because only a small portion of them can be identified through currently available technologies. Furthermore, some agents that we believed to be harmless or kept in-check turned out to be dangerous—how dangerous often isn't known until after a catastrophic event. For this and other reasons, the move to de-prioritize the communicable disease surveillance systems in place in the 1970s was naïve. It was based on the sharp decline of infectious diseases in developed countries,

which was due to the advent of effective but expensive vaccines and antibiotics, and failed to take into account conditions in the developing world. This oversight was especially detrimental in large parts of the developing world, which for many reasons is the main source of emerging infections. But the boomerang effect of the Pyrrhic victory over infectious diseases was felt as a wake-up call by all.

NEW FACTS

Public health experts agree that both the emergence of new infections and the mobilization of resources to keep them under control should be treated as global phenomena; there are no national boundaries for the spread of microbes. However, health care systems operate on a national level. Therefore, a special effort must be made to collect and disseminate health data in order to build an adequate response to a chronic global threat. This state of affairs is evident from two reports published a decade apart by the Institute of Medicine: the 1992 *Emerging Infections: Microbial Threats to Health in the United States* and the 2003 *Microbial Threats to Health: Emergence, Detection, and Response*. Much of the information in this article is derived from the latter report.

Emerging infections are defined as infectious diseases whose incidence in humans has increased in the past two decades or threatens to increase in the near future. These diseases fall into the following categories:

- New infections resulting from changes in or the evolution of existing organisms (Examples: HIV/AIDS; SARS; Nipah virus infections; etc.)
- Known infections spreading to new geographic areas or populations (Examples: West Nile encephalitis; hantavirus pulmonary syndrome; ebola hemorrhagic fever; etc.)
- Previously unrecognized infections appearing in areas undergoing ecologic transformation (Examples: lyme disease; human ehrlichiosis; yellow fever; etc.)
- Old infections reemerging as a result of antimicrobial resistance in known agents or breakdowns in public health measures (Examples: multidrug-resistant tuberculosis; methicillin-resistant *Staphylococcus aureus*; dengue fever and dengue hemorrhagic fever).

The following individual factors have been identified as accountable for emerging infections:

- microbial adaptation and change;
- human vulnerability;
- climate and weather;
- changing ecosystems;
- economic development and land use;

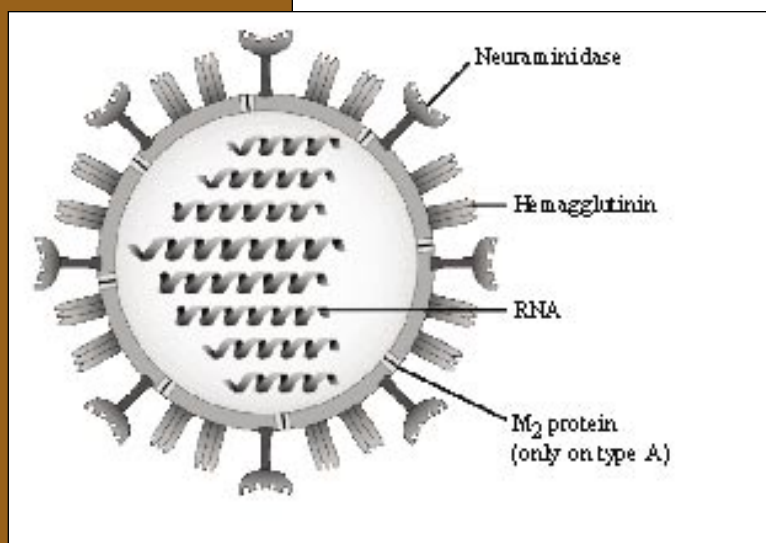


FIGURE 1. Influenza A virus: from American Health Consultants.

- human demographics and behavior;
- technology and industry;
- international travel and commerce;
- breakdown in public health measures;
- poverty and social inequality;
- war and famine;
- lack of political will; and
- intent to harm.

Any of these factors alone can trigger a new disease, but their convergence creates especially high-risk environments, in which infectious diseases can emerge or re-emerge, afflicting individuals and societies alike and posing particular challenges to the medical and public health professionals who must face these situations at the front line.

The national response to the increasing health threats from emerging infections should involve sectors of public health, clinical medicine, and veterinary medicine. Indeed, it has been estimated that 75% of emerging infectious diseases in humans are due to zoonotic pathogens. But of ultimate importance is the understanding that the U.S.'s capacity to respond to microbial threats must include a significant investment in the capacity of developing countries to monitor and address those threats as they arise. It is well established that the greatest burden of infectious diseases today occurs within developing countries, where an estimated one in every two persons dies from such a disease.

Surveillance of infectious diseases is the cornerstone to prevention. One of the key recommendations of the Institute of Medicine is that the U.S. should take a leadership role in promoting a comprehensive system of global surveillance for infectious diseases. At the same time, the U.S. capacity to respond to microbial threats is contingent upon rebuilding the domestic public health infrastructure that has been neglected in the past decades. This will improve domestic surveillance through better disease reporting and exploring innovative systems of surveillance.

The foundation for the above measures, however, lies in research aimed at better understanding the agents that cause infectious diseases. Etiologic diagnosis is the cornerstone of effective disease control and prevention efforts, including surveillance. In the era of large spectrum antibiotics, etiologic diagnosis has been left on the back burner; however, its value in developing new antimicrobials is well recognized.

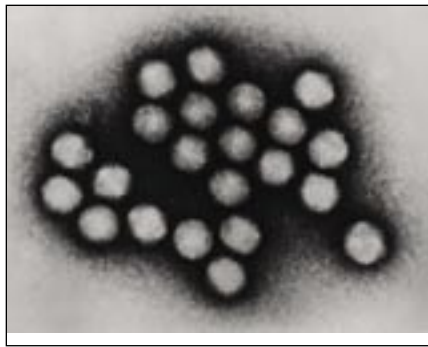


FIGURE 2. Human adenovirus: electron micrograph from *phil.cdc.gov*.

PRACTICE

Academic medical centers are the natural leaders of such research. The University of Iowa responded to the increasing public health demand in 2001 by establishing a Center for Emerging Infectious Diseases (CEID), based in the College of Public Health. The center's strategy is an interdisciplinary approach to research and surveillance. The center employs the latest laboratory technologies, advanced epidemiological methods, and clinical evaluations to study newly emerging pathogens. These studies include international, national, state, and local surveillance for emerging pathogens, risk factor analyses for pathogen infections, vaccine trials, evaluations of rapid diagnostic tests, and, in the future, will include outbreak investigations.

The CEID involves epidemiologists, clinicians, veterinarians, microbiologists, virologists, statisticians, laboratory technologists, laboratory technicians, undergraduate students, and graduate students from The University of Iowa (College of Public Health, University Hygienic Laboratory, and Carver College of Medicine), as well as their collaborators from partnering organizations. CEID investigators are collaborating with national and international experts in adapting and developing laboratory capabilities to detect and study emerging pathogens. Initial efforts are focused upon adenovirus, human metapneumovirus, zoonotic influenza, and West Nile virus infections.

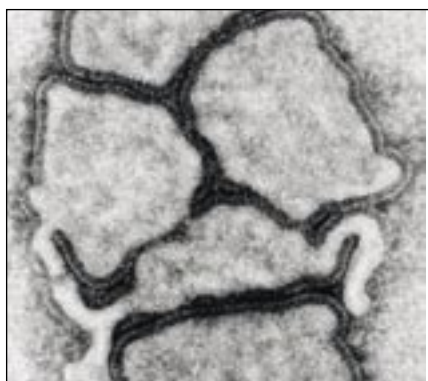


FIGURE 3. Human metapneumovirus: negative-stain electron micrograph, photographed by Charles Humphrey, M.D., of CDC/NCID/IDPA. Published in *JID*, 2002; **185**:1660-3.

The **human adenovirus** is a double-stranded, nonenveloped DNA virus of the family Adenoviridae. There are at least 51 antigenically unique serotypes or species categorized into six subgenera. Recent data suggest that new, possibly more virulent genotypes of these viruses have entered U.S. populations since 1996 and caused at least 26 deaths. Researchers at the CEID have recently demonstrated that one of these new types, adenovirus type 7d2, is now quite common in Iowa and has been associated with recent deaths. Because surveillance for adenovirus in the U.S. is poor, investigators from the CEID and CDC and public health researchers in California and Texas have designed a national surveillance system for emerging adenoviruses that has met with favorable review. Funding is expected next year. The UI College of Public Health CEID investigators will lead this five-year, 20-site surveillance study and guide molecular risk factor studies to determine what makes infection with some of these viruses so dangerous.

Very little is known about the **human metapneumovirus**, a newly described virus of the Paramyxoviridae family. Early data suggest it is genetically most similar to the avian pneumovirus that has caused much illness among domestic turkey flocks. These viruses are single-stranded, negative sense RNA viruses. Early assessments suggest that the human virus may be similar to respiratory syncytial virus in having two distinct genotypes and a similar epidemiology. The virus can cause severe respiratory disease and death; reinfections may occur; and all ages may be susceptible to infections. Recently, in collaborations with investigators in South America, CEID investigators have isolated some of the first human metapneumovirus isolates from the United States and South America. These isolates are being compared molecularly with similar viruses from around the world.

Influenza virus is a pleomorphic, enveloped, single-stranded RNA virus of the family Orthomyxoviridae. There are four subtypes: Genus Influenza virus A, Genus Influenza virus B, Genus Influenza virus C, and Genus "Thogoto-like Viruses." Influenza A and B cause epidemics of respiratory illness in humans and animals. Animal hosts include birds, pigs, ducks, whales, horses, and seals. Variations in the two surface proteins of influenza A, hemagglutinin (H) and neuraminidase (N), are used to describe specific viruses. At least 15 types of hemagglutinin and nine types of neuraminidase are recognized, although not all forms are seen in human infections. Significant change

in these proteins is often associated with epidemics. Worldwide epidemics occur when changes to these proteins are marked, such as when a form of influenza A only seen in animals suddenly begins to infect humans. Human vaccines against influenza A and B are designed annually to protect against the threat of changes in endemic influenza strains.

In recent years, influenza strains normally only detected in birds (H5N1 and H9N2) have caused disease in humans. The cross-species movement of influenza merits close surveillance. In collaboration with investigators from Iowa State University, the University of Wisconsin, and the CDC, researchers from CEID are conducting a number of studies to evaluate the risk of swine and avian influenza viruses infecting humans. Researchers are studying rural Iowans and workers in the meat-packing industry and plan to study veterinarians.

West Nile virus is a single-stranded RNA virus of the family Flaviviridae. It is similar to other arboviruses, such as St. Louis encephalitis virus, LaCrosse virus, and Western Equine encephalitis virus. Common in Africa, the Middle East, and Eastern Europe, West Nile virus first appeared in the U.S. during a 1999 outbreak in New York that killed seven people. Since then, the virus has swept across the country and has now been discovered in 42 states. The virus was first detected in Iowa in September of 2001. The recent emergence of West Nile virus presents a threat to both human and animal health. CEID researchers, led by the University Hygienic Laboratory, are conducting a number of studies of human populations in Iowa for West Nile virus infections.

The CEID works in collaboration with the College of Public Health's Preventive Intervention Center (PIC). The PIC was established in 1992 to conduct large, complex population-based research studies. With its large staff of professionals and enrollment clinics in Iowa City, Des Moines, and Davenport, it is ideally suited for large Phase II-IV vaccine trials.

Via a network of talented infectious disease researchers, the CEID is particularly well-suited to study the epidemiology of emerging diseases, new diagnostics, and new treatments. Many College of Public Health students are interested in its projects, and the expectations are that the CEID will also serve as a training center for tomorrow's infectious disease epidemiologists.

REFERRAL SERVICE

For more information visit the center's website at www.public-health.uiowa.edu/ceid.